



Selected Approximate Approaches to Robust Emergency Service System Design

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Abstract. When the structure of a rescue service system is newly formed or redesigned, several factors possibly affecting the system performance should be taken into account. Changes in the service center deployment following from the results of associated decision-making process are often expected to be applied for a long period and thus, the obtained result has a strategic importance. Usually, the mathematical model of the rescue service system design problem with robustness incorporation takes a complicated structure. This fact disables effective solving of the problem by the branch-and-bound method due to its bad convergence. In this paper, we provide the reader with a short spectrum of our own approximate modelling techniques, which can be used to obtain a sufficient solution of the p -location problem by means of integer programming in acceptably short computational time. Theoretical explanation of suggested methods is here accompanied with the computational study performed with real-sized problem instances.

Keywords: Emergency medical service · p -location problem · Rescue system robustness · Radial formulation · Approximate approaches

1 Introduction

This paper is focused on special class of discrete network location problems, which are solved under uncertainty following from randomly occurring failures in the transportation network, through which the associated service is provided from suitably located service centers [3, 11, 30, 32, 33]. The emergency medical service system design problem is a challenging task for both system designers and operational researchers. As the first ones search for tools, which enable to obtain service center deployment satisfying future demands of the system users in case of emergency, the second ones face the necessity of completing the solving tools. Emergency service system efficiency may be considerably influenced by more or less appropriate locations of the service centers, which send emergency vehicles to satisfy service requests at the system users' locations. The number of service providing centers must be limited due to economic and technological restrictions. Obviously, the number of ambulance vehicles is finite and staff capacities are also restricted [1, 2, 4, 10, 16, 31].

In this paper, we deal with such system design objective, which represents the service accessibility of an average user. Then, the emergency service system design can be tackled as the weighted p -median problem. Any service center is assumed to have enough capacity to serve all assigned users and thus, each system user can be serviced from the nearest located service center. Host of similar models assume that serviced population is concentrated to a finite number of dwelling places of the considered area. Frequency of the demand occurrence at a given place is proportional to the number of inhabitants of the given town or village.

The necessity to solve large problem instances has led to the development of several effective solving techniques. The first group of approaches contains the exact methods based on the branch and bound principle. Many of them use a specific model reformulation, in which so-called radial approach is applied [1, 9, 12, 19]. One of the biggest disadvantages of the exact methods consists in their almost unpredictable demands, which usually make the solving process extremely time consuming. Simultaneously, many different approximate methods have been developed. Currently, the main attention is paid to various metaheuristic approaches, i.e. genetic algorithms, scatter search, path-relinking method and many others [7, 8], the aim of which can be specified as a task of obtaining a good solution in acceptably short computational time.

When the emergency service system is designed, the designer must take into account that the traversing time between a service center and the affected user might be impacted by various random events caused by weather or traffic, and possible failure of a part of critical infrastructure should be taken into account. In other words, the system resistance to such critical events should be included into the decision-making process. Most of available approaches to increasing the system resistance [13, 14, 22, 23] are based on making the design resistant to possible failure scenarios, which can appear in the associated transportation network as a consequence of random failures due to congestion, disruptions or blockages. Thus, a finite set of the failure scenarios is considered and each individual scenario is characterized by particular time distances between the users' locations and possible service center locations.

The most commonly used objective function in the above mentioned weighted p -median problem consists in minimizing the time service accessibility for an average user [16, 28], i.e. min-sum objective function is minimized subject to associated solution feasibility constraints, when the design should be obtained by using common mathematical programming tools.

The system resistance to randomly occurring detrimental events described by the set of scenarios is usually ensured so that the service center deployment has to comply with all specified scenarios. The usual way of taking into account all scenarios is based on minimizing the maximal objective function of the individual instances corresponding with particular scenarios [30, 32]. It means that instead of the min-sum objective function used in the standard weighted p -median problem, the min-max objective function is used here as a design quality level. The min-max link-up constraints and the cardinality of the scenario set significantly change the model structure. They represent an undesirable burden in any integer-programming problem due to bad convergence of the

branch-and-bound method inside most available IP-solvers. Thus, alternative approximate approaches to the scenario based robust design constitute a big challenge to family of operational researchers and professionals in applied informatics.

In this paper, we present several attempts to mastering the complexity of the above scenario based robust design problem and provide the reader with a short spectrum of our own approximate modelling techniques, which can be used to obtain a sufficient solution of the p -location problem by means of integer programming in acceptably short computational time. All suggested and presented approaches are based on so-called radial formulation of the problem, which proved its usefulness in previous research aimed at effective solving of large instances of the weighted p -median problem without the concept of system robustness [9, 12, 19]. Theoretical explanation of suggested methods is here accompanied with the computational study performed with real-sized problem instances.

This book chapter represents an extended version of the paper presented at ICORES 2020 conference. The novelty of this extended paper consists in the content and structure of the paper. The conference paper was aimed directly at only one approximate solving technique for robust service system design problem, which makes use of so-called system reengineering. The original idea of system reengineering comes from previous research reported in [25]. It must be noted that any kind of system robustness was studied in mentioned literature. In the extended paper, we provide the readers with a short spectrum of approaches, which use different assumptions. A fast algorithm suggested in [20] inspired the method proposed in the conference paper, but the inner part of the algorithm was replaced by mentioned reengineering approach. In addition, the sense of system robustness is here discussed. We have added a separate section in which we analyze the possible effects of various adverse events on system performance and service delivery efficiency. Then, a short example is shown to explain the difference between basic and robust service center deployment. Another difference between the papers consists in the fact, that the extended version of the paper consists a preliminary computational study, in which the complexity and time demands of the exact mathematical model are studied. The conference paper di not contain such an analysis. Reported results confirm the necessity of approximate solving approaches development. Some of case studies and tables are taken from our previous research publications, but the reason is to provide the readers with a complex view on system robustness and to bring the possibility of comparing different developed approaches. The mathematical models were developed by the authors for this study, by the inspiration was taken and some partial approaches were taken from previous research. As an example we can mention the reengineering approach, which was developed not for solving the robust service system design problem, but in this study, this concept was incorporated into an effective and fast algorithm.

The remainder of this paper is organized as follows. The next section is devoted to the explanation of system robustness. A short example shows how the unexpected events may influence the resulting service center deployment and why it is important to consider such scenarios in the modelling phase of service system designing. Section 3 contains the exact mathematical model of robust emergency service system design problem. The necessity of approximate solving techniques is here discussed and the computational time demands of the exact model are reported. The fourth section is focused on robustness evaluation possibilities. In the fifth section, we summarize various approximate approaches and provide the reader with a short list of references. Next two sections are devoted to our developed approximate solving approaches. One of them consists in making use of the useful features of the radial formulation in solving the min-sum p -location problem. The second suggested method is based on so-called system reengineering, which considers also current service center deployment and brings only such changes that would be acceptable for service providers. After theoretical explanation of all studied methods, there is an overview of performed numerical experiments and the section yields also brief comparative analysis of designed service center deployments based on different values of reengineering parameters. Finally, the conclusion summarizes obtained findings and contains possible directions of further research.

2 Sense of System Robustness

The main goal of this section is to explain the reasons that led to the idea of rescue system robustness. Generally, when an emergency service system or any other form of public service system is newly formed or redesigned, some standard conditions on the network, through which the service is provided, are usually expected. It means that no unexpected problems or delays in service delivering are assumed. However, this assumption is not correct and it does not hold to all types of service systems. It must be realized that the emergency service system runs continuously 24 h a day, 365 days a year. Logically, many different undesirable effects that may occur randomly anywhere and anytime can negatively affect the system performance. Such critical adverse events represent uncertainty, the source of which can be divided into two main groups:

1. **Endogeneous (Internal) Causes:** These detrimental events come from the service system itself. At the time of requesting the rescue vehicle to depart, the staff may be temporarily unavailable due to the provision of a service to another patient. To incorporate such situations into the decision-making process, the concept of so-called generalized disutility has been introduced [18, 24, 26] and in many others. This model extension enables to cover more demands of system users, which can arise almost simultaneously, and furthermore, the nearest located service center does not need to have sufficient capacity to solve all the assigned demands at the same time. Of course, mentioned model generalization brings some difficulties into the solving process of the problem. Since the concept of generalized disutility does not represent the main research topic of this paper, we will restrict ourselves only on the second group of critical events that may influence the efficiency of service providing.

2. **Exogeneous (External) Causes:** This group of problems includes various adverse events on transportation networks under the influence of daytime (morning and afternoon rush hours), weather, or technical condition of roads. Such events are independent on the designed system, but they can have negative impact on its performance. Such critical events are usually described by separate distance or time matrices and thus, they form so-called detrimental scenarios [30, 32, 34, 35]. Then, the goal of the solving process is to find such service center deployment, which minimizes mentioned negative impact of the detrimental scenarios.

Another aspect that should be taken into account consists is the time perspective. Changes in the service center deployment following from the results of the associated decision-making process are often expected to be applied for a long period and thus, the obtained result has a strategic importance. Therefore, the system robustness should be considered. System robustness can be understood as a task of making the designed system resistant to random detrimental events at the maximal possible level. Let us show a simple practical demonstration.

Simple examples of various factors that may negatively affect the service system operating efficiency and cause heavy traffic on the associated transportation network are shown in Fig. 1.

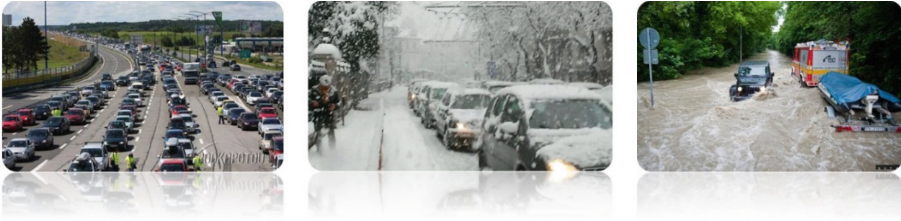


Fig. 1. Uncertainty following from randomly occurring failures in the network [21].

The following practical example of system robustness was originally published in [21]. As it was already mentioned, randomly occurring detrimental events should be considered in the decision making process, i.e., in the strategic phase of the system designing. The following figure shows, how the elongation of one network arc affects the optimal deployment of two service centers. The small numbers placed by the network nodes represent the number of demands for service in particular network nodes. The left part of Fig. 2 represents the standard conditions and the associated optimal deployment of two service centers obtained by solving the weighted 2-median problem. The right part of Fig. 2 contains the same network graph with one elongated arc caused by a detrimental event. We can observe that the depicted arc evaluation significantly changes the optimal solution of the problem and the resulting system designs differ.

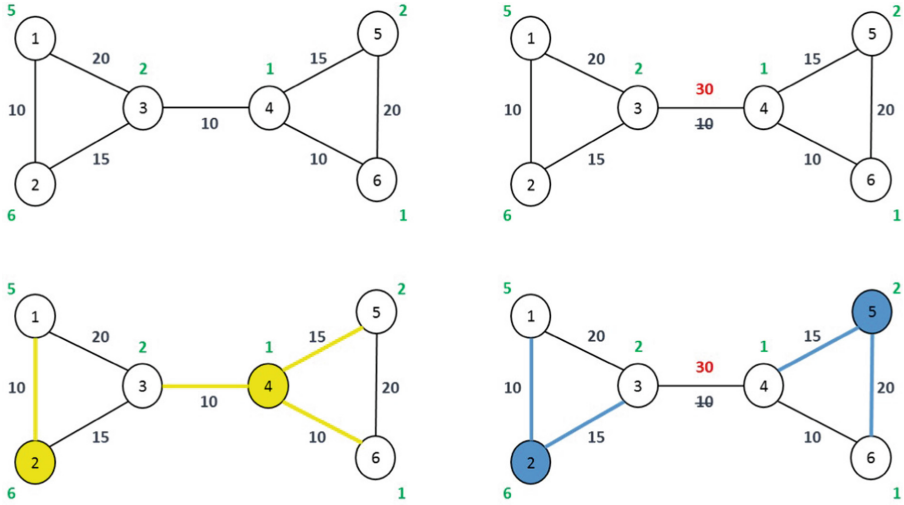


Fig. 2. Impact of a detrimental event on the optimal center deployment [21].

In practical applications of the robust service system design, there are usually more disjoint detrimental events and much bigger networks. Thus, finding the optimal service center deployment concerning system robustness represents a challenging task for the experts in the operations research field.

To incorporate system resistance into the associated mathematical model, a finite detrimental scenario set must be formed to cover the most commonly expected disruptions in service deliveries. This way, the associated public service system can be designed by minimization of the highest detrimental impact of the individual scenarios.

3 Exact Mathematical Model

The standard approach to emergency service system design usually leads to formulation of a min-sum location problem [1, 6, 16, 31], in which the average system accessibility for users (average response time) is minimized. Unfortunately, the robust system design gets more complicated structure and the problem is formulated as a min-max model bringing some difficulties into the computational process [21].

The robust emergency service system design problem using the radial formulation can be described by the following denotations. Let symbol J denote the set of users' locations and let symbol I denote the set of possible service center locations. We denote by b_j the number of users sharing the location j . In some cases, the number b_j could express directly the estimated number of demands for service, but it is generally assumed that the number of ambulance vehicle arrivals to the location j is proportional to the population density in this location. To solve the problem, p locations must be chosen from I so that the maximal scenario objective function value is to be minimal. The objective function value of an individual scenario is defined as a sum of users' distances from the location of the service center providing them with service multiplied by b_j .

Furthermore, let symbol U denote the set of possible failure scenarios. This set contains also one specific scenario called *basic scenario*, which represents standard conditions in the associated transportation network. The integer distance between locations i and j under a specific scenario $u \in U$ is denoted by d_{iju} . Even if the radial model is originally suggested for integer distance or time values only, the used principle enables us to adjust the model also for real values without any big problems.

The decisions, which determine the structure of the rescue service system, are modeled by decision variables y_i for $i \in I$. The variable $y_i \in \{0, 1\}$ models the decision on service center location at the place $i \in I$ by the value of 1 if a service center is located at i and by the value of 0 otherwise. In the robust problem formulation, the variable h as the upper bound of the objective function issues following the individual scenarios is used. To formulate the radial model, the integer range $[0, v]$ of all possible distances of the matrices $\{d_{iju}\}$ is partitioned into zones according to [12, 19]. The value of v is computed according to the expression (1).

$$v = \max\{d_{iju} : i \in I, j \in J, u \in U\} - 1 \quad (1)$$

The zone s corresponds to the interval $(s, s + 1]$. Furthermore, auxiliary zero-one variables x_{jus} for $j \in J, u \in U$ and $s = 0 \dots v$ need to be introduced. The variable x_{jus} takes the value of 1, if the distance of the user at $j \in J$ under the scenario $u \in U$ from the nearest located center is greater than s and it takes the value of 0 otherwise. Then the expression $x_{ju0} + x_{ju1} + x_{ju2} + \dots + x_{juv}$ constitutes the distance d_{ju}^* from user location j to the nearest located service center under scenario $u \in U$. Similarly to the set covering problem, let us introduce a zero-one constant a_{iju}^s under scenario $u \in U$ for each $i \in I, j \in J, s \in [0 \dots v]$. The constant a_{iju}^s is equal to 1, if the disutility d_{iju} between the user location j and the possible center location i is less than or equal to s , otherwise a_{iju}^s is equal to 0. Then the model of the robust system design problem can be formulated as follows.

$$\text{Minimize} \quad h \quad (2)$$

$$\text{Subject to:} \quad x_{jus} + \sum_{i \in I} a_{iju}^s y_i \geq 1 \quad \text{for } j \in J, u \in U, s = 0, 1, \dots, v \quad (3)$$

$$\sum_{i \in I} y_i = p \quad (4)$$

$$\sum_{j \in J} b_j \sum_{s=0}^v x_{jus} \leq h \quad \text{for } u \in U \quad (5)$$

$$y_i \in \{0, 1\} \quad \text{for } i \in I \quad (6)$$

$$x_{jus} \in \{0, 1\} \quad \text{for } j \in J, u \in U, s = 0, 1, \dots, v \quad (7)$$

$$h \geq 0 \quad (8)$$

The objective function (2) represented by single variable h gives an upper bound of all objective function values corresponding to the individual scenarios. The constraints (3) ensure that the variables x_{jus} are allowed to take the value of 0, if there is at least one center located in radius s from the user location j and constraint (4) limits the number of located service centers by p . The link-up constraints (5) ensure that each perceived disutility (time or distance) is less than or equal to the upper bound h . The obligatory constraints (6), (7) and (8) are included to ensure the domain of the decision variables y_i , x_{jus} and h .

While the weighted p -median problem used for simple emergency medical system design without the robustness consideration is easily solvable, the robust design formulated by the model (2)–(8) often causes computational difficulties due to the model size and its complicated structure. Despite the fact that the radial formulation may accelerate the associated solving process [12, 19], such model simplification may not help. Obviously, each additional detrimental scenario rapidly increases the number of decision variables and structural constraints as well. Secondly, the optimization criterion (2) in combination with the link-up constraints (5) considerably slow down the convergence of the branch-and-bound process to the optimal solution. Mentioned weakness was verified in a small computational study, which is reported in the following subsection.

3.1 Complexity of the Exact Model and Its Computational Time Demands

It is natural, that the min-max model (2)–(8) has more complicated structure than the simple weighted p -median problem with a min-sum optimization criterion. That is why we expect that the min-max model size together with its bad structure can cause computational difficulties of the branch-and-bound method, especially with increasing number of detrimental scenarios. To verify this hypothesis, a small portion of preliminary experiments was performed.

To get the optimal solution of the problem (2)–(8), the optimization software FICO Xpress 8.3 (64-bit, release 2016) was used. The parameters of the computer were Intel Core i7 5500U 2.4 GHz processor and 16 GB RAM. As far as the benchmark set is concerned, we took the road network of eight Slovak self-governing regions, in which real emergency health care system is implemented. The set of used problem instances contains the regions of Bratislava (BA), Banská Bystrica (BB), Košice (KE), Nitra (NR), Trenčín (TN), Trnava (TT) and Žilina (ZA). The sets I and J represent all cities and villages of each region. The coefficients b_j were equal to the number of inhabitants rounded up to hundreds. The parameters of individual benchmarks are summarized in the left part of Table 1. The right part of the table contains the exact solutions of the robust rescue system design problem described by the model (2)–(8). The objective function value (2) is reported in the column denoted by h and the computational time in seconds is given in the column denoted by CT [s].

It must be noted that one of the most important aspects of robust service system designing consists in creating an effective set of scenarios making possible to test system robustness regarding the distribution of detrimental events on relevant transportation network graph arcs. Here, many different ways can be used to create the set of scenarios. Despite the easiest random process, a specific analysis of a network graph [5, 15, 17, 27] can be applied to obtain a suitable scenario set.

Due to the lack of common benchmarks for study of robustness, the scenarios used in our computational study were created in the following way. We selected 25 percent of matrix rows so that these rows correspond to the biggest cities concerning the number of system users. Then we chose randomly from 5 to 15 rows from the selected ones and the associated disutility values in the chosen rows were multiplied by the randomly chosen constant from the range 2, 3 and 4. The rows, which were not chosen by this random process, stay unchanged. This way, ten different scenarios were generated for each self-governing region. The scenarios represent the consequence of fatal detrimental events, when some time-distances are several times elongated.

Table 1. Size of tested benchmarks and the exact solutions of the robust system design for self-governing regions of Slovakia with ten detrimental scenarios [20].

Region	$ I $	p	CT [s]	h
BA	87	9	28.7	25417
BB	515	52	1063.1	18549
KE	460	46	1284.2	21286
NR	350	35	2017.6	24193
PO	664	67	1180.4	21298
TN	276	28	264.1	17524
TT	249	25	433.8	20558
ZA	315	32	1229.7	23004

Mentioned demands of the exact model (2)–(8) can be observed also in Fig. 3, which was published in [20]. The figure shows the dependence of average computational time on the cardinality of the scenario set. Obviously, the exact approach does not hold for large problem instances because of their extremely high computational time demands.

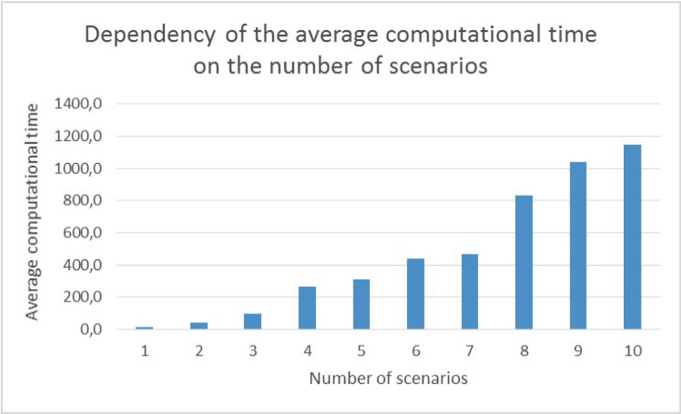


Fig. 3. Dependence of the average computational time on the number of scenarios [20].

As we can see in Table 1 and in Fig. 3, the exact approach to the robust rescue service system design with ten detrimental scenarios is quite time-demanding. Much worse situation occurs, when the cardinality of the scenario set doubles. The following Table 2 contains the results of the same problem instances as before with the only difference consisting in usage of 20 detrimental scenarios for each benchmark instead of 10. Additional scenarios were generated in the same way as before. Another difference is in the computational time, which was limited to 8 h. In the problems, which could not be solved to optimality by given time threshold, the best-found solution h together with the available lower bound LB is reported.

Table 2. Results of numerical experiments for 20 scenarios and 8 h of computational time.

Region	$ I $	p	CT [s]	h	LB
BA	87	9	174.1	27720	
BB	515	52	28810.9	18830	18754
KE	460	46	19239.2	21959	
NR	350	35	28803.3	24584	24537
PO	664	67	28810.1	21603	21556
TN	276	28	15369.8	17775	
TT	249	25	1144.8	21091	
ZA	315	32	4853.9	23230	

Based on reported results, we can conclude that the exact approach to the robust rescue service system design is very time-consuming and thus, approximate solving and modelling techniques need to be developed.

4 Robustness Evaluation

The main goal of robust service system design is to make the system resistant to randomly occurring failures on the associated transportation network. To evaluate the gauges of robustness, several additional denotations are needed. As before, let U denote the set of all considered failure scenarios, which contains also the *basic scenario*. The *basic scenario* corresponds to the standard conditions on the associated network. Let $\mathbf{y}$ denote the vector of location variables y_i ; $i \in I$. Let $\mathbf{y}^b$ correspond to the basic system design, i.e. the solution of a simple weighted p -median problem, in which only the *basic scenario* is taken into account. Let $f^b(\mathbf{y})$ denote the associated objective function value. Similarly, let $\mathbf{y}^r$ denote the solution of the model (2)–(8), which brings the robust system design. Finally, the objective function (2) will be denoted by $f^r(\mathbf{y})$. The *price of robustness (POR)* expresses the relative increment (additional cost) of the basic scenario objective function, when $\mathbf{y}^r$ is applied instead of the optimal solution $\mathbf{y}^b$ obtained for the basic scenario. Its value is defined by (9).

$$POR = \frac{f^b(\mathbf{y}^r) - f^b(\mathbf{y}^b)}{f^b(\mathbf{y}^r)} \quad (9)$$

The *price of robustness* expresses the “bill” for making the system robust, but it does not express what we gain by such application of such solution. Therefore, we introduce also a coefficient called *gain of robustness (GOR)* expressed by (10).

$$GOR = \frac{f^r(\mathbf{y}^b) - f^r(\mathbf{y}^r)}{f^r(\mathbf{y}^r)} \quad (10)$$

This coefficient evaluates the profit following from applying the robust solution instead of the standard one in the worst case ignoring the detrimental scenarios [21].

5 Short Overview of Recent Approximate Approaches

The preliminary computational study with the exact model (2)–(8) reported in Sect. 3.1 confirmed the necessity of various alternative heuristic approaches to be developed. In this section, we provide the reader with a brief overview of possible ways to comply with the model complexity and bad convergence of the branch-and-bound method applied on the model (2)–(8) with the link-up constraints (5) linking the individual scenario objective functions up to their common upper bound h . We present here several approaches based on the radial formulation, which have been recently developed.

The first suggested approach follows the Lagrangean relaxation of the link-up constraints (5). Each of these constraints is associated with a non-negative Lagrangean multiplier λ_u and the sub-gradient method is used to set up suitable values of the multipliers. The original problem (2)–(8) turns into minimization of (11) subject to (2), (4), (6), (7) and (8).

$$\text{Minimize} \quad h + \sum_{u \in U} \lambda_u \left(\sum_{j \in J} \sum_{s=0}^v b_j x_{jsu} - h \right) = h \left(1 - \sum_{u \in U} \lambda_u \right) + \sum_{u \in U} \lambda_u \sum_{j \in J} \sum_{s=0}^v b_j x_{jsu} \quad (11)$$

Obviously, the problem (11), (3), (4), (6), (7) and (8) has a feasible solution only for such setting of Lagrangean multipliers, where their sum is less than or equal to one. Having the optimal solution of the relaxed problem with arbitrary multipliers meeting the stated rule, the value of the optimal solution yields a lower bound of the optimal solution of the original problem (2)–(8). If the sum of the Lagrangean multipliers is less than one, then the optimal value of h would equal to zero and it does not represent the upper bound at all. That is why we restrict ourselves on such setting of multipliers, where the sum equals to one exactly. We try to reach or at least to approximate the optimal solution of the original problem (2)–(8) by an iterative algorithm, where the Lagrangean multipliers λ_u are recomputed in each iteration. This algorithm was introduced in [13], where also other details are discussed.

The rest of presented approaches are based on the idea of replacing the set U of individual scenarios by one supplementary scenario. After this assumption, the original exact min-max model (2)–(8) turns into a simplified to the form (12)–(16).

$$\text{Minimize} \quad \sum_{j \in J} b_j \sum_{s=0}^v x_{js} \quad (12)$$

$$\text{Subject to: } x_{js} + \sum_{i \in I} a_{ij}^s y_i \geq 1 \quad \text{for } j \in J, s = 0, 1, \dots, v \quad (13)$$

$$\sum_{i \in I} y_i = p \quad (14)$$

$$y_i \in \{0, 1\} \quad \text{for } i \in I \quad (15)$$

$$x_{js} \geq 0 \quad \text{for } j \in J, s = 0, 1, \dots, v \quad (16)$$

All decision variables and structural constraints take the same meaning as before. The objective function (12) minimizes the disutility perceived by an average user, what can be achieved by minimizing the sum of time or distance values perceived by all system users. The only difference consists in the fact, that the indexes u over the set U of scenarios used in the variables and constraints of the previous exact mathematical model (2)–(8) are not necessary.

As it was discussed in previous sections, simple weighted p-median problem described by the model (12)–(16) is much better solvable than the original min-max problem formulated by expressions (2)–(8).

Generally, the constant a_{ij}^s for $i \in I, j \in J, s \in [0 \dots v]$ is equal to 1, if the disutility d_{ij} perceived by any user located at the location j from the possible center location i is less than or equal to s , otherwise $a_{ij}^s = 0$. The disutility d_{ij} may be defined in many different ways. It may represent time necessary for service delivering or the distance, which is needed to overcome in order to get the demand satisfied.

The fuzzy approach [14] does not deal with the set of given crisp scenarios as proposed in the model (2)–(8), but it describes the individual uncertain values, e.g. the value of perceived disutility, by a range of possible values together with a measure of relevance of the individual values from the range. The measure is called *membership function* and the function is defined on the whole set of real numbers and maps this definition range on the interval $[0, 1]$. The value of one is assigned to the real values that belong to the fuzzy value at the highest level and the value of zero is intended for real values outside the range of the fuzzy value. A particular fuzzy set defined on real numbers membership function of which it satisfies given conditions is called *fuzzy number*. We restrict ourselves to a special type of the fuzzy sets intended for expressing that a processed uncertain disutility value is greater than or equal to a given real value $d_{ij-\min}$ with tolerance $d_{ij-\max} - d_{ij-\min}$. Here, $d_{ij-\min} = \min\{d_{iju} : u \in U\}$ and $d_{ij-\max} = \max\{d_{iju} : u \in U\}$ (Fig. 4).

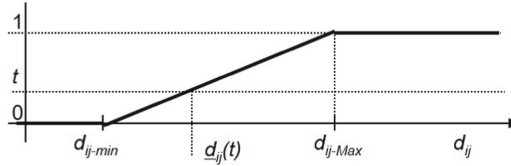


Fig. 4. The membership function of a fuzzy set $\underline{d}_{ij}$.

In this fuzzy approach, the constant a_{ij}^s must be defined for each pair $[i, j]$ and given level of satisfaction $t \in [0, 1]$. The constant $a_{ij}^s(t)$ is equal to 1, if the disutility $\underline{d}_{ij}(t)$ between the user location j and the possible center location i is less than or equal to s , otherwise $a_{ij}^s(t) = 0$. After these preliminaries, we can solve the problem described by the model (12)–(16) with fuzzy coefficients by iterative Tanako-Asai's approach [29, 36], in which we use the bisection method to search for the highest level of satisfaction t^* , for which the problem has a feasible solution. Obviously, one of the problems connected with the fuzzy approach consists in a proper way of fuzzy number computation, which may logically affect the resulting level of satisfaction and the service center deployment as well. The disadvantage of previously introduced approach consists in the fact that it does not process the complete set U in a sufficient way. The minimum and maximum of each disutility over the set U does not bring any information about other scenarios, which should be also considered. Therefore, we have suggested also another approach.

This newly suggested approach is based on a convex combination of individual scenarios. Similarly to the Lagrangean relaxation described above, the set of nonnegative multipliers $\lambda_u \geq 0$ for $u \in U$ is used to define the resulting value of each disutility $\underline{d}_{ij}$ perceived by any user located at j from the possible center location i . It must be noted that the coefficients λ_u have to satisfy the condition that their sum equals to one. These multipliers can also represent the probability distribution of the individual scenarios. Then, the disutility values $\underline{d}_{ij}$ can be expressed according to (17) as follows.

$$\underline{d}_{ij}(\lambda) = \sum_{u \in U} \lambda_u d_{iju} \quad \text{for } i \in I, j \in J \quad (17)$$

The values of d_{ij} depend on the settings of multipliers λ_u . After these preliminaries, the covering constants a_{ij}^s are computed and the radial model (12)–(16) is iteratively solved. In each iteration, the multipliers λ_u are redefined based on the obtained solution according to certain rules [22, 23]. The radial model is being solved, while better solution of the system design keeps being obtained or until other stopping criteria are met.

The main advantage of suggested heuristic approaches consists in the fact that the approximate methods perform in order faster than the original standard exact approach, which is formulated by the model (2)–(8). The difference is caused by the fact, that the approximate approaches do not process all the detrimental scenarios and avoid introducing the min-max link-up constraints, which generally cause bad convergence of the branch-and-bound method.

As far as the solution accuracy is concerned, the quality of obtained results is usually strongly dependent on the initial settings of parameters or on the way of fuzzy numbers definition. If we had to choose the best heuristic method out of mentioned approaches, we would pick the convex approach, because it takes into account the complete set U of scenarios and does not reflect only to some selected values of disutility. Therefore, we will concentrate our effort on development of such algorithm that would find a good solution in a short time without introducing specific parameters.

6 Fast Algorithm with Fixed Location Variables

The main reason for developing approximate solving approaches to robust rescue system design problem is to overcome bad solvability of the original problem described by the model (2)–(8) caused by its structure and big size. It must be realized that each additional detrimental scenario magnifies the model size proportionally to the cardinality of the set I .

Presented algorithm is based on the assumption that not only the cardinality of the set U , link-up constraints (5) and the min-max objective function (2), but also the number of location variables y_i play an important role and make the model extremely time-consuming for the associated solving method. The main idea of this effective and fast algorithm consists in the following expectation. If we were able to make the decision problem smaller by fixing some variables y_i either to the value 0 or 1, then we could considerably accelerate the solving process. Thus, the original exact model can be replaced by series of smaller sub-problems to find suitable variables y_i for fixing. After that, we can solve a smaller min-max robust rescue system design problem. Mentioned adjustment turns the original exact approach to an approximate solving technique. To formulate a new mathematical model with mentioned adjustment, three new disjoint sets must be defined in such a way, that their union brings the original set of all possible service center locations I . Let the set F_1 contain such indexes $i \in I$, for which y_i must equal to 1, i.e. a service center must be located in each element of F_1 . Analogically, the set F_0 is formed by those center locations, in which a center cannot be located. Here, $y_i = 0$ for each $i \in F_0$. Finally, let the indexes of variables y_i , which do not have assigned their values yet, form a finite set denoted by V . Mathematically, the set V can be defined as $V = I - F_1 - F_0$. Furthermore, it is enough to define variables y_i only for the locations from the set V , because the decisions of other service center locations are already made

by the sets F_0 and F_1 . Based on these preliminaries, the obligatory constraints (6) can be replaced by (18) and the constraints (4) can take the form of (19). In the expression (19), the symbol p_{rem} is used to denote the number of service centers, which remain to the original number of all located service centers p . It means, $p_{rem} = p - |F_1|$. Obviously, it is assumed that the set F_1 contains less than p elements.

$$y_i \in \{0, 1\} \quad \text{for } i \in V \quad (18)$$

$$\sum_{i \in V} y_i = p_{rem} \quad (19)$$

Similarly, it is not necessary to define the coefficient a_{iju}^s for each $i \in I$ as above. It is enough to define it for each element i from $V \cup F_1$. Then, the constraints (3) can be redefined to (20).

$$x_{jus} + \sum_{i \in V} a_{iju}^s y_i + \sum_{i \in F_1} a_{iju}^s \geq 1 \quad \text{for } j \in J, u \in U, s = 0, 1, \dots, v \quad (20)$$

After mentioned model adjustment suggestions, the approximate radial model for robust rescue system design takes the form of (21)–(27). The meaning of decision variables stays unchanged.

$$\text{Minimize} \quad h \quad (21)$$

$$\text{Subject to: } x_{jus} + \sum_{i \in V} a_{iju}^s y_i + \sum_{i \in F_1} a_{iju}^s \geq 1 \quad \text{for } j \in J, u \in U, s = 0, 1, \dots, v \quad (22)$$

$$\sum_{i \in V} y_i = p_{rem} \quad (23)$$

$$\sum_{j \in J} b_j \sum_{s=0}^v x_{jus} \leq h \quad \text{for } u \in U \quad (24)$$

$$y_i \in \{0, 1\} \quad \text{for } i \in V \quad (25)$$

$$x_{jus} \in \{0, 1\} \quad \text{for } j \in J, u \in U, s = 0, 1, \dots, v \quad (26)$$

$$h \geq 0 \quad (27)$$

Filling up the sets F_1 and F_0 can be done in the following way. Let us introduce a set I_1^u for each detrimental scenario $u \in U$. The set I_1^u contains the solution of a simple weighted p -median problem obtained for each scenario u separately. The set I_1^u for given $u \in U$ can be mathematically formulated by (28).

$$I_1^u = \{i \in I : y_i = 1\} \quad \text{for } u \in U \quad (28)$$

The mentioned location problem solved for each scenario $u \in U$ to define the set I_1^u can be formulated either by the model (2)–(8), in which the scenario set U contains only the scenario u , or we can solve a simple weighted p -median problem making use of the above introduced constants and variables. The radial formulation for a given scenario u takes the form of (12)–(16). In the model (12)–(16), we work only with one scenario u from the scenario set U and therefore, we do not use the index u in variables x_{js} .

As shown in [12, 19], this model can be solved to optimality in a very short time thank to good characteristics of the covering problem. After computing the set I_1^u for each $u \in U$, we can fill the sets F_1 and F_0 according to (29) and (30) respectively.

$$F_1 = \bigcap_{u \in U} I_1^u \quad (29)$$

$$F_0 = I - \bigcup_{u \in U} I_1^u \quad (30)$$

Finally, the complete approximate algorithm for rescue system designing can be summarized according to [20] by the following steps:

1. Determine the set I_1^u for each detrimental scenario $u \in U$ by solving the model (12)–(16).
2. Fill up the sets F_1 and F_0 according to (29) and (30).
3. Solve the model (21)–(27) to obtain the resulting system design.

Suggested approximate algorithm can be easily implemented in any common optimization environment without the necessity of special software tool development. It is based on two integer programming problems, which can be solved by the branch-and-bound or similar method. Another big advantage of presented approach consists in its independence on any parameters, which are often sensitive to their values and may directly affect not only the performance characteristics (computational time), but also the obtained results. Thus, this approximate solving method can be considered very useful.

7 Reengineering Approach

The problem of many optimization approaches applied on real systems consists in the fact that the resulting service center deployment may bring too big changes in service center deployment, which would be hardly acceptable for service providers. To avoid such potential complications, the concept of system reengineering has been introduced [24–26]. Its most important feature lies in considering current service center deployment and performing only acceptable relocations satisfying given restrictions. In the approximate robust system design algorithm suggested in Sect. 6, the reengineering concept can be incorporated into the solving process in the way of filling the sets F_0 and F_1 respectively [25]. Let us now remind the basis of system reengineering.

The basic idea follows from the analysis of current service center deployment, which may not be optimal due to changing demands and development of the underlying transportation network. To explain the problem in more details, consider the simple example depicted in Fig. 5. We assume that the left graph represents current deployment of four service centers marked by blue color. All the vertices represent possible demand points. To evaluate the current center deployment, the sum of distances from each network node to the nearest located center was used as the quality criterion. Here, it takes the value of 66. If we allowed changes in current service center locations and moved a service center from the node 2 to the node 6, we would perform system reengineering and we could achieve better value of the criterion. The new system design is depicted on the right graph and its evaluation is 64. By this small example, we demonstrated the principle and goal of system reengineering.

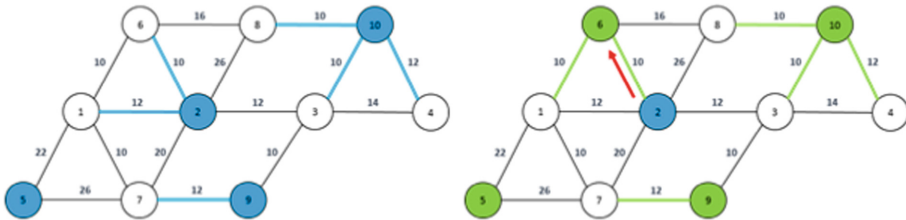


Fig. 5. Simple example of emergency service system reengineering [26].

To describe the problem of the system average distance or response time minimization by changing the deployment of centers, let I be a finite set of all possible center locations. As above, the response time following from the distance between locations i and j under a specific scenario $u \in U$ is denoted as d_{iju} . The union of two disjoint sets of located centers L and F_1 , where I_L contains p centers under reconstruction and F_1 is the set of fixed centers, describe the current emergency service center deployment. The center locations from I_L can be relocated within the set $I - F_1 - F_0$, where F_0 is the set of temporarily forbidden locations.

When reengineering of an emergency service system is performed, the administrator of the system may set up parameters of rules to prevent a designer of new center deployment from changes, which can be considered by system users as obnoxious. We consider two formal rules within this study. The first rule limits the total number w of the centers, location of which can be changed. The second rule limits the time distance between current and newly suggested location of a service center by the given value D . This rule can be explained also by the following example depicted in Fig. 6. We assume that all points 1–11 represent system users and the black points 2, 3, 9 and 11 represent current service center locations.

To be able to formulate the rules in a concise way, we derive several auxiliary structures. Let N_t denote the set of all possible center locations, to which the center $t \in I_L$ can be moved subject to limited length of the move. The set N_t can be mathematically formulated by the expression (31).

$$N_t = \{i \in I - F_1 - F_0 : d_{ti} \leq D\} \quad \text{for } t \in I_L \quad (31)$$

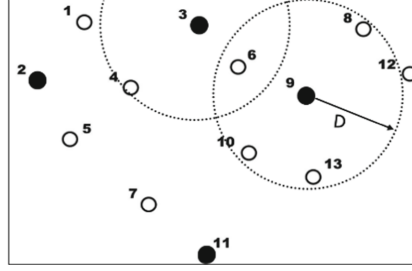


Fig. 6. Simple example of reengineering restrictions [26].

If we consider the example depicted in Fig. 6, we can observe that the center located at the point 9 can be moved to 6, 8, 10 and 13 or stay unchanged. Thus, the set $N_9 = \{6, 8, 9, 10, 13\}$. Similarly, $N_3 = \{3, 4, 6\}$.

Additionally, let S_i denote the set of all centers, which can be moved to $i \in I - F_1 - F_0$ subject to the mentioned limitation as described by (32). Here, $S_6 = \{3, 9\}$.

$$S_i = \{t \in I_L : i \in N_t\} \quad \text{for } i \in I - F_1 - F_0 \quad (32)$$

Now, we introduce series of decision reallocation variables, which model the decisions on moving centers from their original positions to new ones. The variable $u_{ti} \in \{0, 1\}$ for $t \in L$ and $i \in N_t$ takes the value of one, if the service center at t is to be moved to i and it takes the value of zero otherwise. The following model (33)–(41) describes the problem of reengineering for given detrimental scenario u .

$$\text{Minimize} \quad \sum_{j \in J} b_j \sum_{s=0}^v x_{js} \quad (33)$$

$$\text{Subject to:} \quad x_{js} + \sum_{i \in I - F_1 - F_0} a_{iju}^s y_i + \sum_{i \in F_1} a_{iju}^s \geq 1 \quad \text{for } j \in J, s = 0, 1, \dots, v \quad (34)$$

$$\sum_{i \in I - F_1 - F_0} y_i = p \quad (35)$$

$$\sum_{i \in L} y_i \geq p - w \quad (36)$$

$$\sum_{i \in N_t} u_{ti} = 1 \quad \text{for } t \in L \quad (37)$$

$$\sum_{t \in S_i} u_{ti} \leq y_i \quad \text{for } i \in I - F_1 - F_0 \quad (38)$$

$$u_{ti} \in \{0, 1\} \quad \text{for } t \in L, i \in N_t \quad (39)$$

$$y_i \in \{0, 1\} \quad \text{for } i \in I - F_1 - F_0 \quad (40)$$

$$x_{js} \geq 0 \text{ for } j \in J, s = 0, 1, \dots, v \quad (41)$$

As the constraints (34), (35) and formulae together with used decision variables were explained in the previous section, we restrict the explanation only on the remainder of the above model. Constraint (36) limits the number of changed center locations by the constant w . The next constraints (37) allow moving the center from the current location t to at most one other possible location in the radius D . Constraints (38) enable to bring at most one center to a location i subject to the condition that the original location of the brought center lies in the radius D . These constraints also assure consistency among the decisions on move and decisions on center location. The constraints (39), (40) and (41) are used to keep the definition scope of used variables.

Having introduced and explained the service system reengineering problem, this model can be incorporated into the approximate solving algorithm for robust service system designing. If we set the parameters w and D of the reengineering method at chosen values from the ranges $[1 \dots p]$ and $[1 \dots D^{max}]$ respectively, and initialize the sets $F_1 = \emptyset$ and $F_0 = \emptyset$, then the following steps can describe the suggested solving algorithm:

1. Determine the set I_1^u for each detrimental scenario $u \in U$ by solving the reengineering model (33)–(41) with the parameters w and D .
2. Fill up the sets F_1 and F_0 according to (29) and (30).
3. Solve the model (21)–(27) to obtain the resulting system design.

Let the indexes $i \in I - F_1 - F_0$, for which the location variable y_i in the model (21)–(27) take the value of 1 together with the indexes i from the set F_1 form the resulting service system deployment described by the set I_{res} as defined by (42).

$$I_{res} = F_1 \cup \{i \in I - F_1 - F_0 : y_i = 1\} \quad (42)$$

The objective function value h of any resulting set I_{res} can be computed according to the expression (43).

$$h = f(I_{res}) = \max \left\{ \sum_{j \in J} b_j \min \{d_{iju} : i \in I_{res}\} : u \in U \right\} \quad (43)$$

8 Computational Study

8.1 Benchmarks and Preliminaries

The numerical experiments reported in this case study were performed in the optimization software FICO Xpress 8.3, 64-bit. The experiments were run on a PC equipped with the Intel® Core™ i7 5500U 2.4 GHz processor and 16 GB RAM.

Used benchmarks were derived from real emergency health care system, which was implemented in eight regions of Slovak Republic. For each self-governing region, i.e.

Bratislava (BA), Banská Bystrica (BB), Košice (KE), Nitra (NR), Prešov (PO), Trenčín (TN), Trnava (TT) and Žilina (ZA), all cities and villages with corresponding number of inhabitants b_j were taken into account. The coefficients b_j were rounded up to hundreds. The set of communities represents both the set J of users' locations and the set I of possible center locations as well. The cardinalities of these sets vary from 87 to 664 locations. The organization of the Slovak self-governing regions is depicted in Fig. 7.



Fig. 7. Used benchmarks – self-governing regions of Slovakia.

The parameters of individual benchmarks are summarized in the Table 1 in Sect. 3.1, in which also the exact solutions taken from previous research [21] are reported. For this computational study, 10 scenarios were generated according to the algorithm making use of random generation. The main goal of this computational study was to verify basic performance characteristics of approaches suggested in Sects. 6 and 7.

The solution accuracy is evaluated by so-called *gap*, which is defined in the following way. Let $ObjF_{ex}$ denote the objective function value of the exact solution and let us use $ObjF_{ap}$ to denote the objective function value of the approximate one. Then we can evaluate *gap* in percentage of $ObjF_{ex}$ according to the following expression (44).

$$gap = 100 * \frac{ObjF_{ap} - ObjF_{ex}}{ObjF_{ex}} \quad (44)$$

The resulting system designs are compared by *Hamming distance* of the location variables vectors. Generally, *Hamming distance* HD of two vectors $\mathbf{y}^a$ and $\mathbf{y}^b$ is defined by the expression (45). Obviously, the vectors must have the same size.

$$HD(\mathbf{y}^a, \mathbf{y}^b) = \sum_{i \in I} (y_i^a - y_i^b)^2 \quad (45)$$

8.2 Experiments with the Fast Algorithm with Fixed Location Variables

This chapter contains the results of performed computational study, the goal of which was to verify basic performance characteristics of suggested fast algorithm with fixed location variables. As discussed in previous sections, the exact approach to robust rescue

system design formulated by the model (2)–(8) is very time-demanding and thus, approximate solving and modelling techniques need to be developed. The results obtained by performed numerical experiments are reported in the following Table 3.

Each row of the table corresponds to one solved problem instance. The objective function value computed for the resulting system design according to the expression (43) is reported in the column denoted by h and the computational time in seconds is given in the column denoted by CT [s]. Furthermore, the solution accuracy is evaluated by so-called *gap*, which is defined in the form of (44). Finally, the resulting system designs are compared by *Hamming distance* of the location variables vectors. These values are reported in the last column of the table denoted by HD .

Table 3. Results of the approximate approach for the self-governing regions of Slovakia [20].

Region	$ I $	p	Approximate solution			
			CT [s]	h	gap [%]	HD
BA	87	9	19.2	25417	0.00	0
BB	515	52	84.3	18779	1.24	18
KE	460	46	66.4	21664	1.78	14
NR	350	35	65.0	24392	0.82	12
PO	664	67	129.6	21584	1.34	18
TN	276	28	62.7	17744	1.19	10
TT	249	25	29.5	21084	2.56	12
ZA	315	32	24.8	23359	1.54	10

The results reported in Table 3 clearly confirm expected useful features of suggested approximate algorithm, because it provides us with very satisfactory solutions from the viewpoint of accuracy. Furthermore, the demanded computational time is in order smaller than required by the exact approach. When analyzing the values of location variables y_i for $i \in I$, we have found the assumption that obtaining better solution may be probably limited by the set F_1 , which contains also such indexes $i \in I$ of variables y_i fixed to 1, that should belong to the set V instead of F_1 . The second portion of numerical experiments verified this hypothesis. Here, the set F_1 keeps empty and the approximate algorithm uses only the set F_0 . In such a case, we can expect these two consequences:

1. Obtaining better result from the viewpoint of objective function value (43), because there are less location variables fixed to their final values and thus, the solution accuracy may get better.
2. Higher computational time demands, because there are more variables, which need to get their final values.

The following portion of experiments, results of which are reported in Table 4, confirmed all mentioned expected consequences. This table has the same structure as previously reported Table 3.

Table 4. Results of the approximate approach applied on the self-governing regions, in which only set F_0 is used [20].

Region	$ I $	p	Approximate solution			
			CT [s]	h	gap [%]	HD
BA	87	9	13.5	25417	0.00	0
BB	515	52	203.0	18596	0.25	12
KE	460	46	179.1	21341	0.26	4
NR	350	35	319.7	24193	0.00	0
PO	664	67	216.0	21311	0.06	6
TN	276	28	103.1	17628	0.53	8
TT	249	25	83.6	20768	1.2	4
ZA	315	32	142.8	23004	0.00	0

As we can see in Table 4, presented approximate algorithm, which uses only the set F_0 , brings excellent quality of the resulting solution, which is near to the optimal one. Another useful feature consists in computational time, which keeps satisfactorily small.

8.3 Experiments with the Reengineering Approach

The last but not least important portion of numerical experiments was focused on the approximate solving technique described in Sect. 7, which subsequently solves the system reengineering problem for each detrimental scenario as proposed in [25].

The following series of experiments was performed with the goal to find a suitable setting of the parameters w and D used in the approximate approach, see the model (33)–(41) of the reengineering process. For this study, the benchmark Žilina (ZA) was used. In this portion of experiments, the parameter p was set at the value 32 reported in previous tables. The maximal radius D was fixed at one of the values 5, 10, 15, 20 and 25 and the maximal number w of centers allowed to change their locations was set to $p/4$, $p/2$, $3p/4$, and p respectively. The results of this series of experiments are summarized in Table 5.

Each row of the table corresponds to one setting of the parameters w and D . In this portion of experiments, several characteristics were studied. The computational time in seconds is reported in the column denoted by CT [s]. Since the approximate approach proved to be much faster than the exact one, the percentage save of computational time SCT was computed. Here, the computational time of the exact approach was taken as the base. Furthermore, the objective function associated with the obtained service center deployment is reported in the column denoted by Obj^{ap} . Its value was computed

according to (43). To evaluate the accuracy of suggested approximate method, the value of *gap* was also computed. It expresses the difference between the objective function values of the exact and approximate models. The objective value of the exact approach was taken as the base. The value of *gap* is reported also in percentage. Finally, the resulting service center deployments were compared in the terms of *Hamming distance* of the vectors of location variables y_i . This value is denoted by *HD*.

It can be seen that the lowest computational time of the approximate method was reached for the settings $w = p/4$ and $D = 5$.

Table 5. Detailed results of numerical experiments for the self-governing region of Žilina: computational study of the impact of individual parameters on the results accuracy [25].

w	D	CT [s]	SCT [%]	$ObjF^{ap}$	gap [%]	HD
8	5	231.6	82.25	23411	1.77	14
8	10	430.9	66.98	23377	1.62	14
8	15	502.8	61.46	23236	1.01	6
8	20	516.8	60.39	23359	1.54	10
8	25	529.5	59.42	23359	1.54	10
16	5	234.8	82.00	23411	1.77	14
16	10	455.1	65.12	23377	1.62	14
16	15	516.0	60.45	23236	1.01	6
16	20	529.9	59.38	23359	1.54	10
16	25	542.6	58.41	23359	1.54	10
24	5	232.5	82.18	23411	1.77	14
24	10	434.6	66.69	23377	1.62	14
24	15	508.5	61.02	23236	1.01	6
24	20	531.3	59.28	23359	1.54	10
24	25	534.6	59.02	23359	1.54	10
32	5	231.6	82.25	23411	1.77	14
32	10	427.8	67.21	23377	1.62	14
32	15	509.2	60.97	23236	1.01	6
32	20	531.4	59.27	23359	1.54	10
32	25	534.8	59.01	23359	1.54	10

As the associated *gap* was acceptable, we used this setting in the third series of experiments, which was performed for each self-governing region. The obtained results are reported in Table 6, in which the same denotations as in previous tables were used. The table is divided into two sections denoted by **EXACT** and **APPROXIMATE**. The first section contains the results from Table 1 for bigger comfort of the readers. The second section contains the results obtained by the approximate approach.

Table 6. Comparison of the approximate approach with the system reengineering to the exact approach. Parameters of the approximate approach were set in this way: $w = p/4$, $D = 5$ [25].

	EXACT		APPROXIMATE			
	CT [s]	$ObjF^{ex}$	CT [s]	$ObjF^{ap}$	gap [%]	HD
BA	52.8	25417	11.8	26197	3.07	4
BB	1605.0	18549	679.8	18861	1.68	12
KE	1235.5	21286	633.4	21935	3.05	16
NR	11055.1	24193	274.2	24732	2.23	14
PO	3078.2	21298	1601.9	21843	2.56	20
TN	616.6	17535	223.1	17851	1.80	10
TT	563.8	20558	152.4	20980	2.05	10
ZA	1304.7	23004	231.6	23411	1.77	14

The approximate approach enables to obtain the resulting robust service center deployment in the computational time, which is much less than half of the computational time demanded by the exact approach. As far as the accuracy of the resulting solution is concerned, it can be observed that the approximate method is very satisfactory.

9 Conclusions

This paper was focused on advanced way of rescue service system designing. Emergency medical system plays a very important role in human life. It must be realized that any potentially bad decision made in the process of ambulance vehicle stations locating may bring not only enormous economic losses, but it can cause serious lasting consequences for human health and thus, it usually negatively affects the most vulnerable part of population. Therefore, it is necessary to consider all aspects that may have a bad impact on service delivering and efficiency of system performance.

Within this study, we paid attention to system robustness, which means its resistance to randomly occurring failures in the associated transportation network used by ambulance vehicles when delivering rescue service to system users. To incorporate system robustness into the particular integer-programming model, a finite set of failure scenarios is usually formed. Then, the optimal robust rescue system design can be obtained by minimization of the highest detrimental impact of the individual scenarios.

Since the exact mathematical model has a complicated structure, which causes bad convergence of the branch-and-bound method, several approximate approaches need to be developed. In this paper, we provided the readers with a short spectrum of our own approximate modelling techniques, which can be used to obtain a sufficient solution of the p -location problem by means of integer programming in acceptably short computational time. Special attention was paid to a specific model adjustment consisting in fixing some location variables and in making the decision-making problem smaller. We have suggested two different variants of mentioned approximate algorithm. One of

them consists in making use of the useful features of the radial formulation in solving the min-sum p -location problem. The second suggested method is based on system reengineering, which considers also current service center deployment and brings only such changes that would be acceptable for service providers. Theoretical explanation of all suggested methods was accompanied with the computational study performed with real-sized problem instances.

The performed numerical experiments have confirmed the usefulness of suggested methods. They are simply implementable in any commercial IP-solver without the necessity of special knowledge in the Applied Informatics field. Based on achieved results we can conclude that we have presented very useful tools for robust system designing.

In the future, we would like to concentrate on other approximate techniques, which would enable us to reach shorter computational time under acceptable solution accuracy. Another research goal could be focused on developing such a heuristic approach, which would form the second stage of presented algorithm and could improve the obtained resulting system design. Furthermore, we could focus on goal programming approaches to robust service system design or on developing advanced solving techniques based on evolutionary or machine learning methods.

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References

1. Avella, P., Sassano, A., Vasil'ev, I.: Computational study of large scale p -median problems. *Math. Program.* **109**(1), 89–114 (2007)
2. Brotcorne, L., Laporte, G., Semet, F.: Ambulance location and relocation models. *Eur. J. Oper. Res.* **147**, 451–463 (2003)
3. Correia, I., Saldanha da Gama, F.: Facility locations under uncertainty, *Location Science*, eds. Laporte, Nickel, Saldanha da Gama, pp. 177–203 (2015)
4. Current, J., Daskin, M., Schilling, D.: Discrete network location models. In: Drezner, Z., et al. (eds.) *Facility Location Applications and Theory*, pp. 81–118. Springer, Berlin (2002)
5. Czimmermann, P., Koháni, M.: Characteristics of changes of transportation performance for pairs of critical edges. *Commun. Sci. Lett. Univ. Žilina* **20**(3), 84–87 (2018)
6. Czimmermann, P., Koháni, M.: Computation of transportation performance in public service systems. In: *IEEE Workshop on Complexity in Engineering - COMPENG 2018*, pp. 1–5. Institute of Electrical and Electronics Engineers, Danvers (2018)
7. Doerner, K.F., et al.: Heuristic solution of an extended double-coverage ambulance location problem for Austria. *CEJOR* **13**(4), 325–340 (2005)
8. Elloumi, S., Labbé, M., Pochet, Y.: A new formulation and resolution method for the p -center problem. *INFORMS J. Comput.* **16**, 84–94 (2004)
9. García, S., Labbé, M., Marín, A.: Solving large p -median problems with a radius formulation. *INFORMS J. Comput.* **23**(4), 546–556 (2011)

10. Guerriero, F., Miglionico, G., Olivito, F.: Location and reorganization problems: the Calabrian health care system case. *Eur. J. Oper. Res.* **250**(3), 939–954 (2016)
11. Ingolfsson, A., Budge, S., Erkut, E.: Optimal ambulance location with random delays and travel times. *Health Care Manag. Sci.* **11**(3), 262–274 (2008)
12. Janáček, J.: Approximate covering models of location problems. In: *Proceedings of the 1st International Conference ICAOR. Lecture Notes in Management Science*, Yerevan, Armenia, pp. 53–61 (2008)
13. Janáček, J., Kvet, M.: Designing a robust emergency service system by Lagrangean relaxation. In: *Mathematical Methods in Economics 2016, Liberec*, pp. 349–353 (2016)
14. Janáček, J., Kvet, M.: An approach to uncertainty via scenarios and fuzzy values. *Croatian Oper. Res. Rev.* **8**(1), 237–248 (2017)
15. Janáček, J., Kvet, M.: Detrimental scenario construction based on network link characteristics. In: *Proceedings of the 19th IEEE International Carpathian Control Conference (ICCC)*, Szilvásvárad, pp. 629–632 (2018)
16. Jánošíková, Ľ.: Emergency medical service planning. *Commun. Sci. Lett. Univ. Žilina* **9**(2), 64–68 (2007)
17. Jenelius, E.: Network structure and travel patterns: explaining the geographical disparities of road network vulnerability. *J. Transp. Geogr.* **17**, 234–244 (2009)
18. Kvet, M.: Computational study of radial approach to public service system design with generalized utility. In: *Digital Technologies 2014: Proceedings of the 10th International IEEE Conference*, pp. 198–208 (2014)
19. Kvet, M.: Advanced radial approach to resource location problems. In: Rocha, Á., Reis, L.P. (eds.) *Developments and Advances in Intelligent Systems and Applications. SCI*, vol. 718, pp. 29–48. Springer, Cham (2018). https://doi.org/10.1007/978-3-319-58965-7_3
20. Kvet, M.: Fast approximate algorithm for robust emergency system design. In: *20th International Carpathian Control Conference, Krakow*, pp. 1–6 (2019)
21. Kvet, M.: Complexity and scenario robust service system design. In: *Information and Digital Technologies 2019: Conference Proceedings*, pp. 271–274 (2019)
22. Kvet, M., Janáček, J.: Hill-Climbing algorithm for robust emergency system design with return preventing constraints. *Contemp. Issues Econ.* **2017**, 156–165 (2017)
23. Kvet, M., Janáček, J.: Struggle with curse of dimensionality in robust emergency system design. In: *Mathematical Methods in Economics 2017*, pp. 396–401 (2017)
24. Kvet, M., Janáček, J.: Reengineering of the emergency service system under generalized disutility. In: *7th International Conference on Operations Research and Enterprise Systems, ICORES 2018, Madeira, Portugal*, pp. 85–93 (2018)
25. Kvet, M., Janáček, J.: Robust emergency system design using reengineering approach. In: *9th International Conference on Operations Research and Enterprise Systems, ICORES 2020, Valletta, Malta*, pp. 172–178 (2020)
26. Kvet, M., Janáček, J., Kvet, M.: Computational study of emergency service system reengineering under generalized disutility. In: Parlier, G.H., Liberatore, F., Demange, M. (eds.) *ICORES 2018. CCIS*, vol. 966, pp. 198–219. Springer, Cham (2019). https://doi.org/10.1007/978-3-030-16035-7_11
27. Majer, T., Palúch, S.: Rescue system resistance to failures in transport network. In: *Proceedings of 34th International Conference Mathematical Methods in Economics, Liberec: Technical University of Liberec*, pp. 518–522 (2016)
28. Marianov, V., Serra, D.: Location problems in the public sector. In: Drezner, Z. (ed.) *Facility Location - Applications and Theory*, pp. 119–150. Springer, Berlin (2002)
29. Ramík, J., Vlach, M.: Generalized Concavity in Fuzzy Optimization and Decision Analysis, p. 296. Kluwer Academic Publishers, Boston (2002)
30. Pan, Y., Du, Y., Wei, Z.: Reliable facility system design subject to edge failures. *Am. J. Oper. Res.* **4**, 164–172 (2014)

31. Reuter-Oppermann, M., van den Berg, P.L., Vile, J.L.: Logistics for emergency medical service systems. *Health Syst.* **6**(3), 187–208 (2017)
32. Scaparra, M.P., Church, R.L.: Location Problems under Disaster Events. Location Science, eds. Laporte, Nikel, Saldanha da Gama (2015)
33. Schneeberger, K., Doerner, K.F., Kurz, A., Schilde, M.: Ambulance location and relocation models in a crisis. *CEJOR* **24**(1), 1–27 (2014). <https://doi.org/10.1007/s10100-014-0358-3>
34. Snyder, L.V., Daskin, M.S.: Reliability models for facility location; the expected failure cost case. *Transp. Sci.* **39**(3), 400–416 (2005)
35. Sullivan, J.L., Novak, D.C., Aultman-Hall, L., Scott, D.M.: Identifying critical road segments and measuring system-wide robustness in transportation networks with isolating links: a link-based capacity-reduction problem. *Transp. Res. Part A* **44**, 323–336 (2010)
36. Teodorović, D., Vukadinović, K.: Traffic Control and Transport Planning: A Fuzzy Sets and Neural Networks Approach. Kluwer Academic Publishers, Boston (1998)